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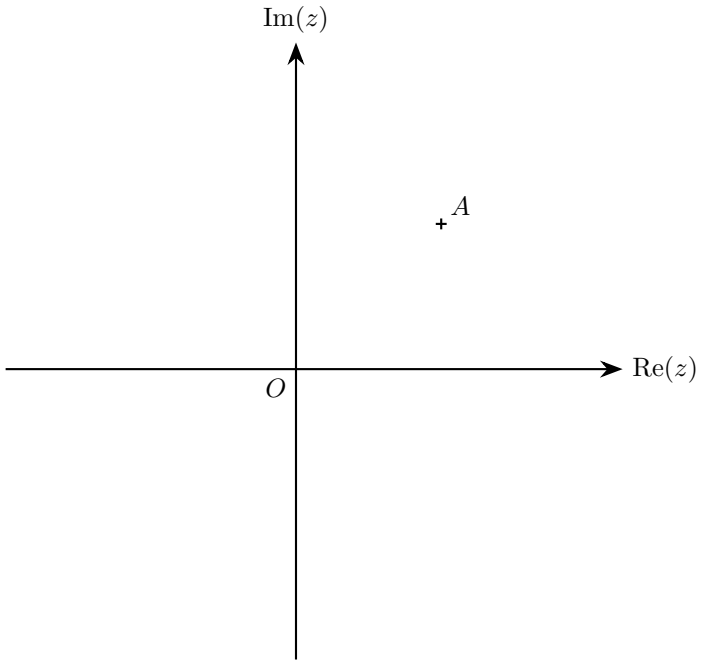


1. An Argand diagram with the point A representing a complex number z_1 is shown in the diagram. The complex numbers z_2, z_3 and z_4 are $z_1 e^{\frac{1}{2}\pi \mathrm{i}}$, $z_1 e^{\pi \mathrm{i}}$ and $z_1 e^{\frac{3}{2}\pi \mathrm{i}}$ respectively.

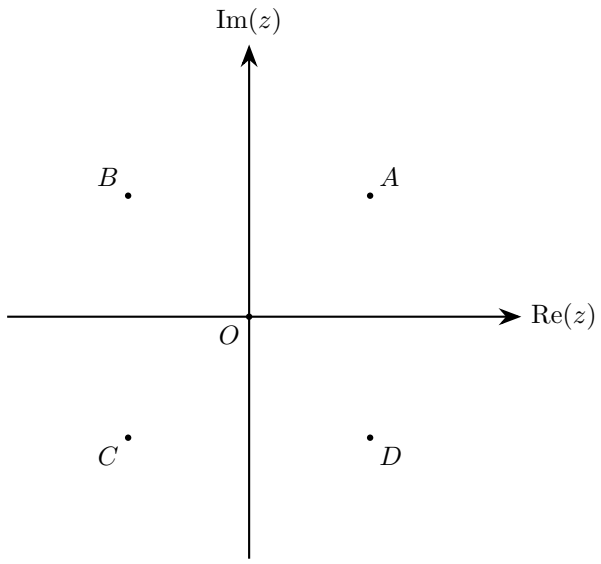
(a) (i) On a copy of the Argand diagram, mark the points B, C and D representing the complex numbers z_2, z_3 and z_4. **[2]**

(ii) Show that $z_1 + z_2 + z_3 + z_4 = 0$. **[2]**

(b) Given now that z_1, z_2, z_3 and z_4 are roots of the equation $z^4 = -16$, find these four roots, giving your answers in the form $a + \mathrm{i}b$, where a and b are real and exact. **[4]**

Solution

(a) (i) Multiplying z_1 by $e^{i\theta}$ corresponds to rotating z_1 by θ anticlockwise about the origin. So B, C and D are the images of A under such rotations of $90°, 180°$ and $270°$ respectively.



(ii)

$$\begin{aligned} z_1 + z_2 + z_3 + z_4 &= z_1 + z_1 e^{\frac{1}{2}\pi i} + z_1 e^{\pi \mathrm{i}} + z_1 e^{\frac{3}{2}\pi i} \\ &= z_1 + \mathrm{i} z_1 - z_1 - \mathrm{i} z_1 \\ &= z_1(1 + \mathrm{i} - 1 - \mathrm{i}) \\ &= 0 \end{aligned}$$

Hence $z_1 + z_2 + z_3 + z_4 = 0$.

(Alternatively, use that $1 + e^{\frac{1}{2}\pi i} + e^{\pi \mathrm{i}} + e^{\frac{3}{2}\pi i}$ is a geometric sum with common ratio $r = e^{\frac{1}{2}\pi i}$)

(b) Write

$$\begin{aligned} -16 &= 16 e^{\mathrm{i}\pi} \\ &= 16 e^{\mathrm{i}(\pi + 2k\pi)} \end{aligned}$$

so the fourth roots are

$$z = 2e^{\mathrm{i}\left(\frac{\pi + 2k\pi}{4}\right)} \qquad k = 0, 1, 2, 3$$

Hence

$$\begin{aligned} k = 0: &\quad z = 2e^{\mathrm{i}\pi/4} = 2\left(\frac{\sqrt{2}}{2} + \mathrm{i}\frac{\sqrt{2}}{2}\right) = \sqrt{2} + \mathrm{i}\sqrt{2} \\ k = 1: &\quad z = 2e^{3\mathrm{i}\pi/4} = 2\left(-\frac{\sqrt{2}}{2} + \mathrm{i}\frac{\sqrt{2}}{2}\right) = -\sqrt{2} + \mathrm{i}\sqrt{2} \\ k = 2: &\quad z = 2e^{5\mathrm{i}\pi/4} = 2\left(-\frac{\sqrt{2}}{2} - \mathrm{i}\frac{\sqrt{2}}{2}\right) = -\sqrt{2} - \mathrm{i}\sqrt{2} \\ k = 3: &\quad z = 2e^{7\mathrm{i}\pi/4} = 2\left(\frac{\sqrt{2}}{2} - \mathrm{i}\frac{\sqrt{2}}{2}\right) = \sqrt{2} - \mathrm{i}\sqrt{2} \end{aligned}$$

Therefore the four roots are

$$\sqrt{2} + \mathrm{i}\sqrt{2}, \quad -\sqrt{2} + \mathrm{i}\sqrt{2}, \quad -\sqrt{2} - \mathrm{i}\sqrt{2}, \quad \sqrt{2} - \mathrm{i}\sqrt{2}$$

2. The five solutions of the equation $z^5 = 16 + 16\sqrt{3}\,\mathrm{i}$ are represented by points on an Argand diagram.

These points are the vertices of a regular pentagon P with centre at the origin.

(a) Determine the five complex numbers that represent these vertices, giving your answers in the form $re^{\mathrm{i}\theta}$, where $0 \le \theta < 2\pi$. **[4]**

(b) Sketch P on an Argand Diagram, taking care to label each vertex. **[2]**

(c) Show that the area of P can be written $k \sin \frac{2\pi}{5}$ where k is an integer to be determined. **[4]**

Solution

(a) Let

$$w = 16 + 16\sqrt{3}\,\mathrm{i}$$

First write w in polar form.
Its modulus is

$$\begin{aligned} |w| &= \sqrt{16^2 + (16\sqrt{3})^2} \\ &= \sqrt{256 + 768} \\ &= \sqrt{1024} \\ &= 32 \end{aligned}$$

Its argument is

$$\arg(w) = \tan^{-1}\left(\frac{16\sqrt{3}}{16}\right) = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$$

since w lies in the first quadrant.
So

$$16 + 16\sqrt{3}\,\mathrm{i} = 32e^{\mathrm{i}\pi/3}$$

Now solve

$$z^5 = 32e^{\mathrm{i}\pi/3}$$

The fifth roots have modulus

$$32^{1/5} = 2$$

and arguments

$$\theta = \frac{\frac{\pi}{3} + 2\pi k}{5} = \frac{\pi}{15} + \frac{2\pi k}{5} \qquad (k = 0, 1, 2, 3, 4)$$

Hence the five roots are

$$\begin{aligned} z &= 2e^{\mathrm{i}\pi/15} \\ z &= 2e^{\mathrm{i}7\pi/15} \\ z &= 2e^{\mathrm{i}13\pi/15} \\ z &= 2e^{\mathrm{i}19\pi/15} \\ z &= 2e^{\mathrm{i}25\pi/15} = 2e^{\mathrm{i}5\pi/3} \end{aligned}$$

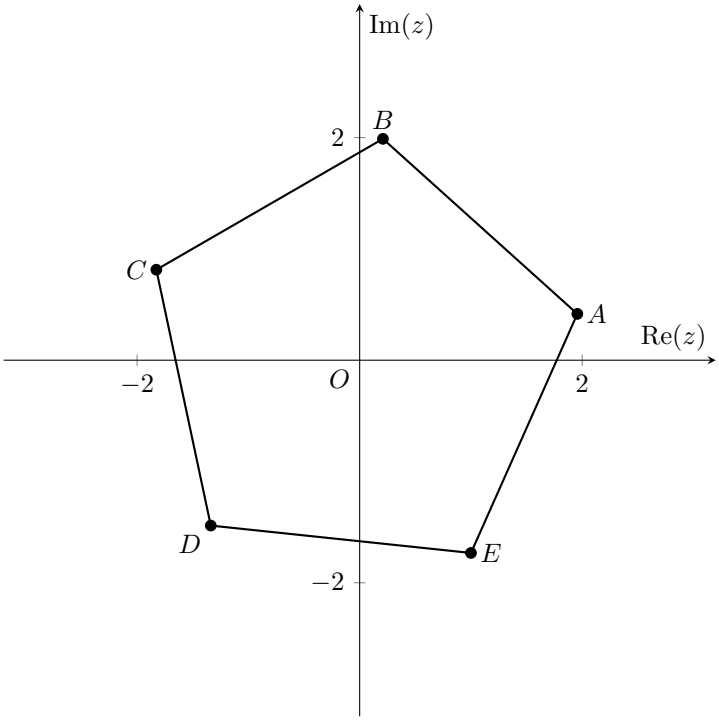
So the five vertices are

$$2e^{\mathrm{i}\pi/15}, \quad 2e^{\mathrm{i}7\pi/15}, \quad 2e^{\mathrm{i}13\pi/15}, \quad 2e^{\mathrm{i}19\pi/15}, \quad 2e^{\mathrm{i}5\pi/3}$$

(b) Let the vertices in anticlockwise order be

$$A = 2e^{\mathrm{i}\pi/15}, \quad B = 2e^{\mathrm{i}7\pi/15}, \quad C = 2e^{\mathrm{i}13\pi/15}, \quad D = 2e^{\mathrm{i}19\pi/15}, \quad E = 2e^{\mathrm{i}5\pi/3}$$

They all lie on the circle of radius 2 centred at the origin, with equal angular spacing $\frac{2\pi}{5}$.



(c) Join the origin to each vertex. This splits the pentagon into 5 congruent isosceles triangles.
Each vertex has modulus 2, so each triangle has two sides of length 2.
Also, the arguments of adjacent vertices differ by

$$\frac{2\pi}{5}$$

so the angle at the origin in each triangle is $\frac{2\pi}{5}$.
Hence the area of one triangle is

$$\frac{1}{2} \times 2 \times 2 \times \sin\frac{2\pi}{5} = 2\sin\frac{2\pi}{5}$$

Therefore the area of the pentagon P is

$$5\left(2\sin\frac{2\pi}{5}\right) = 10\sin\frac{2\pi}{5}$$

So it can be written in the form $k\sin\frac{2\pi}{5}$ with

$$k = 10$$

Hence the area of P is

$$10\sin\frac{2\pi}{5}$$

3. (i) Find the 8th roots of $-128 + 128\sqrt{3}\,\mathrm{i}$ in the form $re^{\mathrm{i}\theta}$, where $r > 0$ and $0 \le \theta < 2\pi$. **[5]**

(ii) These 8th roots are represented in the Argand diagram, in order of increasing θ, by the points A, B, C, D, E, F, G, H.
Draw the Argand diagram, making clear which point is which. **[2]**

(iii) The mid-point of BD is the point P which represents the complex number w.
Find, in exact form, the modulus and argument of w. **[3]**

(iv) w is an nth root of a real number a, where n is a positive integer. State the least possible value of n and find the corresponding value of a. **[3]**

Solution

(i) Let

$$z = -128 + 128\sqrt{3}\,\mathrm{i}$$

Its modulus is

$$\begin{aligned} |z| &= \sqrt{(-128)^2 + (128\sqrt{3})^2} \\ &= 128\sqrt{1+3} \\ &= 256 \end{aligned}$$

Also,

$$\tan\theta = \frac{128\sqrt{3}}{-128} = -\sqrt{3}$$

Since z lies in the second quadrant, its argument is

$$\theta = \frac{2\pi}{3}$$

So

$$-128 + 128\sqrt{3}\,\mathrm{i} = 256e^{\mathrm{i}2\pi/3}$$

If z_k is an 8th root, then

$$z_k^8 = 256e^{\mathrm{i}2\pi/3}$$

Hence the modulus of each root is

$$256^{1/8} = 2$$

and the arguments are

$$\frac{\frac{2\pi}{3} + 2k\pi}{8} = \frac{\pi}{12} + \frac{k\pi}{4}, \qquad k = 0, 1, \ldots, 7$$

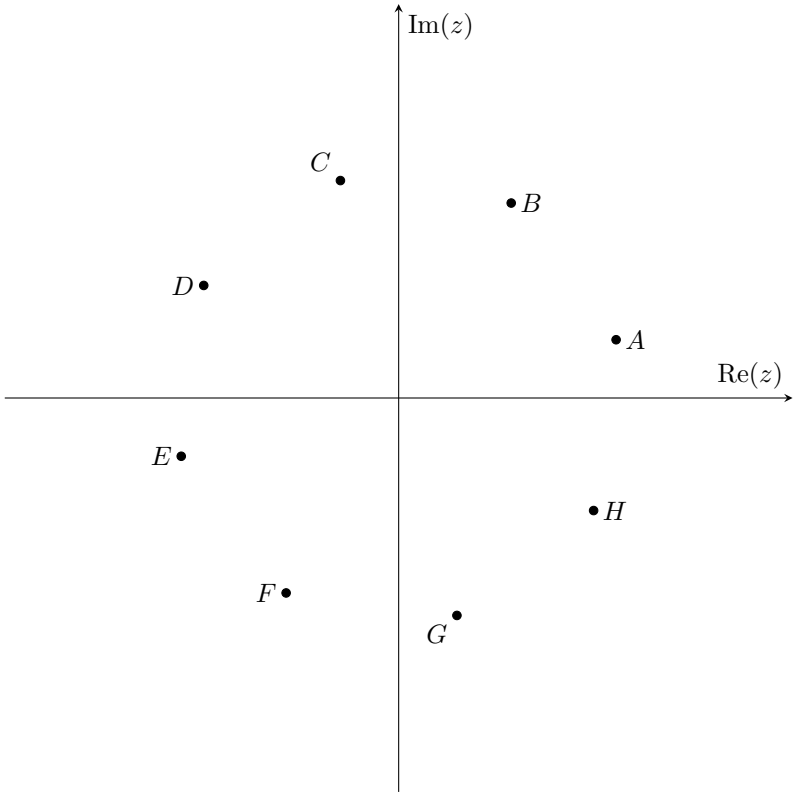
Therefore the 8th roots are

$$z_k = 2e^{\mathrm{i}(\pi/12 + k\pi/4)}, \qquad k = 0, 1, \ldots, 7$$

(ii) In increasing order of argument, the points are

$$A : \frac{\pi}{12}, \quad B : \frac{\pi}{3}, \quad C : \frac{7\pi}{12}, \quad D : \frac{5\pi}{6}, \quad E : \frac{13\pi}{12}, \quad F : \frac{4\pi}{3}, \quad G : \frac{19\pi}{12}, \quad H : \frac{11\pi}{6}$$

All eight points lie on the circle $|z| = 2$.



(iii) The complex numbers represented by B and D are

$$2e^{\mathrm{i}\pi/3} \quad \text{and} \quad 2e^{\mathrm{i}5\pi/6}$$

Since P is the midpoint of BD, the complex number w is

$$\begin{aligned} w &= \frac{2e^{\mathrm{i}\pi/3} + 2e^{\mathrm{i}5\pi/6}}{2} \\ &= e^{\mathrm{i}\pi/3} + e^{\mathrm{i}5\pi/6} \\ &= e^{\mathrm{i}7\pi/12}\left(e^{-\mathrm{i}\pi/4} + e^{\mathrm{i}\pi/4}\right) \\ &= e^{\mathrm{i}7\pi/12}\left(2\cos\frac{\pi}{4}\right) \\ &= \sqrt{2}\, e^{\mathrm{i}7\pi/12} \end{aligned}$$

So the modulus and argument of w are

$$|w| = \sqrt{2}, \qquad \arg(w) = \frac{7\pi}{12}$$

(iv) Since

$$w = \sqrt{2}\, e^{\mathrm{i}7\pi/12}$$

we have

$$w^n = (\sqrt{2})^n e^{\mathrm{i}7n\pi/12}$$

For w^n to be real, its argument must be a multiple of π, so

$$\frac{7n\pi}{12} = m\pi$$

for some integer m. Hence

$$\frac{7n}{12} \in \mathbb{Z}$$

As 7 and 12 have no common factor, the least possible value is

$$n = 12$$

Then

$$\begin{aligned} a &= w^{12} \\ &= (\sqrt{2})^{12} e^{\mathrm{i}7\pi} \\ &= 2^6(-1) \\ &= -64 \end{aligned}$$

So the least possible value of n is 12, and the corresponding value of a is -64.

4. Given that $z_1 = 7 - \mathrm{i}$ and $z_2 = \dfrac{z_1}{1-\mathrm{i}}$,

(a) find z_2 in the form $a + b\mathrm{i}$, where a and b are real. **[2]**

(b) Show on an Argand diagram the point P representing z_1 and the point Q representing z_2. **[2]**

(c) Given that O is the origin, show that $\angle OQP = \dfrac{\pi}{2}$. **[2]**

The circle passing through the points O, P and Q has centre C. Find

(d) the complex number represented by C. **[2]**

(e) the exact value of the radius of the circle. **[2]**

Solution

(a) Rationalise the denominator:

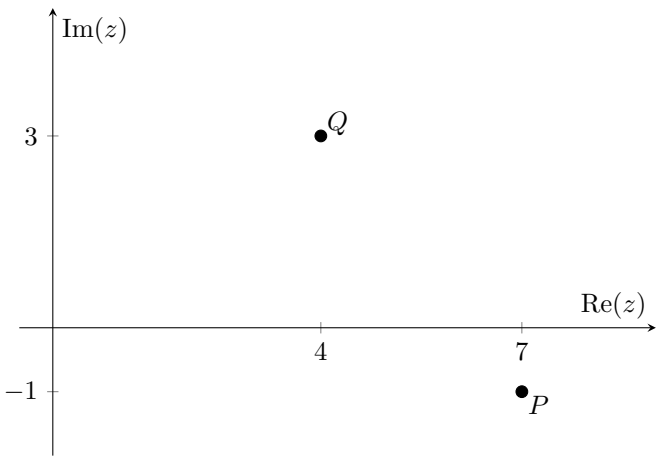
$$\begin{aligned} z_2 &= \frac{7-\mathrm{i}}{1-\mathrm{i}} \cdot \frac{1+\mathrm{i}}{1+\mathrm{i}} \\ &= \frac{(7-\mathrm{i})(1+\mathrm{i})}{(1-\mathrm{i})(1+\mathrm{i})} \\ &= \frac{7+7\mathrm{i}-\mathrm{i}-\mathrm{i}^2}{1-\mathrm{i}^2} \\ &= \frac{7+6\mathrm{i}+1}{1+1} \\ &= \frac{8+6\mathrm{i}}{2} \\ &= 4+3\mathrm{i} \end{aligned}$$

Hence,

$$z_2 = 4 + 3\mathrm{i}$$

(b) On an Argand diagram, the complex number $x + y\mathrm{i}$ is represented by the point (x, y). So

$$z_1 = 7 - \mathrm{i} \;\Rightarrow\; P(7, -1), \qquad z_2 = 4 + 3\mathrm{i} \;\Rightarrow\; Q(4, 3)$$



(c) Using the coordinates from part (b),

$$O = (0, 0), \quad P = (7, -1), \quad Q = (4, 3)$$

Take the vectors from Q:

$$\overrightarrow{QO} = (-4, -3), \qquad \overrightarrow{QP} = (3, -4)$$

Their scalar product is

$$\begin{aligned}\overrightarrow{QO}\cdot\overrightarrow{QP} &= (-4)(3)+(-3)(-4)\\ &= -12+12\\ &= 0\end{aligned}$$

So $\overrightarrow{QO}$ is perpendicular to $\overrightarrow{QP}$, which means

$$\angle OQP = \frac{\pi}{2}$$

(d) Since $\angle OQP = \frac{\pi}{2}$, the side OP is a diameter of the circle through O, P and Q.

Therefore the centre C is the midpoint of $O(0,0)$ and $P(7,-1)$:

$$\begin{aligned}C &= \left(\frac{0+7}{2}, \frac{0+(-1)}{2}\right)\\ &= \left(\frac{7}{2}, -\frac{1}{2}\right)\end{aligned}$$

So the complex number represented by C is

$$\frac{7}{2} - \frac{1}{2}\mathrm{i}$$

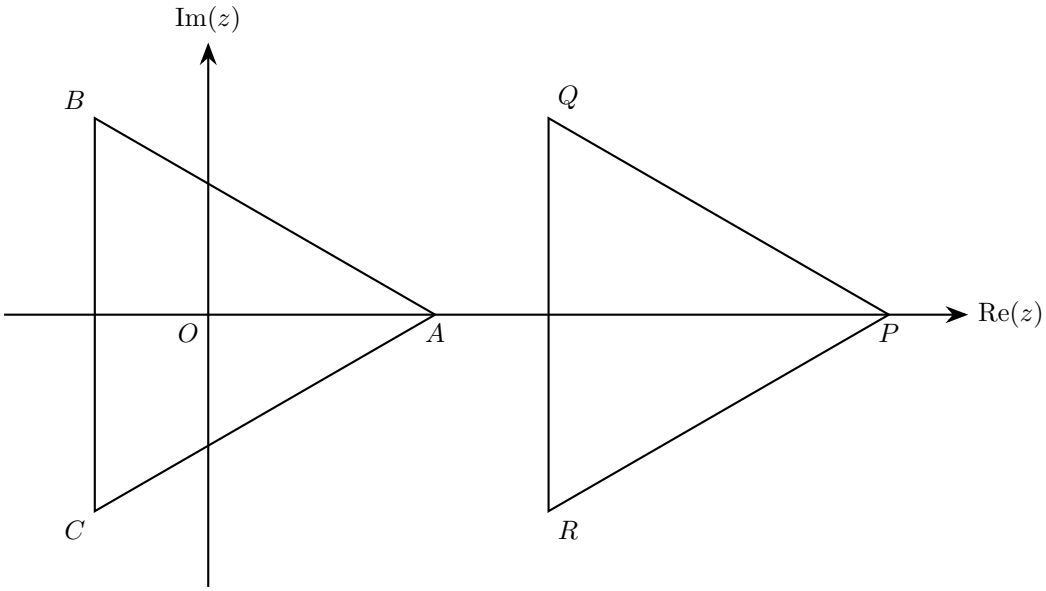
(e) The diameter is OP, so first find its length:

$$\begin{aligned}OP &= \sqrt{(7-0)^2+(-1-0)^2}\\ &= \sqrt{49+1}\\ &= \sqrt{50}\\ &= 5\sqrt{2}\end{aligned}$$

Hence the radius of the circle is

$$\frac{1}{2}OP = \frac{5\sqrt{2}}{2}$$

So the exact radius is $\frac{5\sqrt{2}}{2}$.



5. The cube roots of unity are represented on the Argand diagram below by the points A, B and C.

The complex numbers represented by P, Q and R are obtained by adding 2 to the complex numbers represented by A, B and C respectively.

Determine a degree 6 polynomial equation with integer coefficients whose roots are the complex numbers represented by the points A, B, C, P, Q and R. **[5]**

Solution

Let the required polynomial have variable z.
The points A, B, C represent the cube roots of unity, so their complex numbers satisfy

$$z^3 = 1$$

Hence A, B, C are the roots of

$$z^3 - 1 = 0$$

Now P, Q, R are obtained by adding 2 to the complex numbers at A, B, C.
So if z represents one of P, Q, R, then $z - 2$ is a cube root of unity. Therefore

$$(z-2)^3 = 1$$

so

$$(z-2)^3 - 1 = 0$$

Expanding:

$$\begin{aligned}(z-2)^3 - 1 &= (z^3 - 6z^2 + 12z - 8) - 1 \\ &= z^3 - 6z^2 + 12z - 9\end{aligned}$$

So P, Q, R are the roots of

$$z^3 - 6z^2 + 12z - 9 = 0$$

Therefore a degree 6 polynomial with roots A, B, C, P, Q, R is

$$(z^3 - 1)(z^3 - 6z^2 + 12z - 9) = 0$$

Now expand:

$$\begin{aligned}(z^3 - 1)(z^3 - 6z^2 + 12z - 9) &= z^3(z^3 - 6z^2 + 12z - 9) - (z^3 - 6z^2 + 12z - 9) \\ &= z^6 - 6z^5 + 12z^4 - 9z^3 - z^3 + 6z^2 - 12z + 9 \\ &= z^6 - 6z^5 + 12z^4 - 10z^3 + 6z^2 - 12z + 9\end{aligned}$$

Hence the required polynomial equation is

$$z^6 - 6z^5 + 12z^4 - 10z^3 + 6z^2 - 12z + 9 = 0$$

6. In an Argand diagram, the points A, B and C are the vertices of an equilateral triangle with centre G. The point G represents the complex number $1-2\mathrm{i}$, and the point A represents the complex number $5+2\mathrm{i}$.

(a) Find the complex numbers represented by the points B and C, giving your answers in the form $x + \mathrm{i}y$, where x and y are real and exact. **[6]**

The points D, E and F are the centroids of triangles GBC, GCA and GAB respectively.

(b) Find the exact area of triangle DEF. **[4]**

Solution

(a) Let

$$z_G = 1 - 2i, \qquad z_A = 5 + 2i$$

Then the vector from G to A is

$$z_A - z_G = (5 + 2i) - (1 - 2i) = 4 + 4i$$

In an equilateral triangle, the vectors from the centre to the three vertices are $120°$ apart. So the vectors from G to B and C are obtained by rotating $4 + 4i$ through $\pm 120°$.

Take B to be the $+120°$ rotation. Then

$$z_B - z_G = (4 + 4i)\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)$$

Expanding,

$$\begin{aligned} z_B - z_G &= 4\left(-\frac{1}{2}\right) + 4\left(\frac{\sqrt{3}}{2}i\right) + 4i\left(-\frac{1}{2}\right) + 4i\left(\frac{\sqrt{3}}{2}i\right) \\ &= -2 + 2\sqrt{3}\,i - 2i + 2\sqrt{3}\,i^2 \\ &= -2 - 2\sqrt{3} + (2\sqrt{3} - 2)i \end{aligned}$$

Hence

$$\begin{aligned} z_B &= z_G + (z_B - z_G) \\ &= (1 - 2i) + \left[-2 - 2\sqrt{3} + (2\sqrt{3} - 2)i\right] \\ &= -1 - 2\sqrt{3} + (2\sqrt{3} - 4)i \end{aligned}$$

Now take C to be the $-120°$ rotation. Then

$$z_C - z_G = (4 + 4i)\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)$$

Expanding,

$$\begin{aligned} z_C - z_G &= 4\left(-\frac{1}{2}\right) + 4\left(-\frac{\sqrt{3}}{2}i\right) + 4i\left(-\frac{1}{2}\right) + 4i\left(-\frac{\sqrt{3}}{2}i\right) \\ &= -2 - 2\sqrt{3}\,i - 2i - 2\sqrt{3}\,i^2 \\ &= -2 + 2\sqrt{3} - (2 + 2\sqrt{3})i \end{aligned}$$

Hence

$$\begin{aligned} z_C &= z_G + (z_C - z_G) \\ &= (1 - 2i) + \left[-2 + 2\sqrt{3} - (2 + 2\sqrt{3})i\right] \\ &= -1 + 2\sqrt{3} - (4 + 2\sqrt{3})i \end{aligned}$$

Therefore

$$B = -1 - 2\sqrt{3} + (2\sqrt{3} - 4)i, \qquad C = -1 + 2\sqrt{3} - (4 + 2\sqrt{3})i$$

with B and C interchangeable.

(b) Since G is the centre of the equilateral triangle ABC, it is also the centroid of ABC. Therefore

$$z_G = \frac{z_A + z_B + z_C}{3}$$

so

$$z_B + z_C = 3z_G - z_A$$

The centroid of triangle GBC is D, so

$$\begin{aligned} z_D &= \frac{z_G + z_B + z_C}{3} \\ &= \frac{z_G + (3z_G - z_A)}{3} \\ &= \frac{4z_G - z_A}{3} \end{aligned}$$

Similarly,

$$z_E = \frac{4z_G - z_B}{3}, \qquad z_F = \frac{4z_G - z_C}{3}$$

Now compare these with z_G:

$$\begin{aligned} z_D - z_G &= \frac{4z_G - z_A}{3} - z_G = \frac{z_G - z_A}{3} = -\frac{1}{3}(z_A - z_G) \\ z_E - z_G &= -\frac{1}{3}(z_B - z_G) \\ z_F - z_G &= -\frac{1}{3}(z_C - z_G) \end{aligned}$$

So triangle DEF is similar to triangle ABC, with all lengths scaled by a factor of $\frac{1}{3}$. Hence the area scale factor is

$$\left(\frac{1}{3}\right)^2 = \frac{1}{9}$$

Next, find the area of ABC

First,

$$GA = |z_A - z_G| = |4 + 4i| = \sqrt{4^2 + 4^2} = 4\sqrt{2}$$

Since G is the centre of the equilateral triangle, the three angles at G are equal and sum to $360°$, so

$$\angle AGB = 120°$$

Also,

$$GA = GB = GC = 4\sqrt{2}$$

Hence

$$\begin{aligned} \text{Area of } \triangle AGB &= \frac{1}{2}(GA)(GB)\sin 120° \\ &= \frac{1}{2}(4\sqrt{2})(4\sqrt{2}) \cdot \frac{\sqrt{3}}{2} \\ &= 16 \cdot \frac{\sqrt{3}}{2} \\ &= 8\sqrt{3} \end{aligned}$$

The triangle ABC is made up of the three congruent triangles AGB, BGC and CGA, so

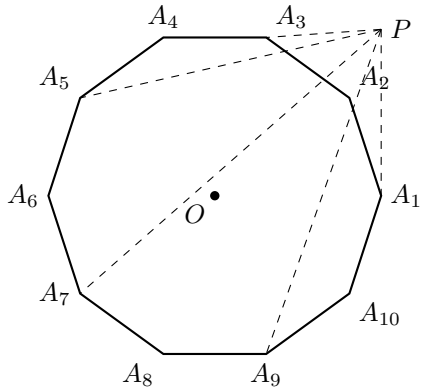
$$\text{Area of } ABC = 3(8\sqrt{3}) = 24\sqrt{3}$$

Therefore

$$\begin{aligned}\text{Area of } DEF &= \frac{1}{9} \times 24\sqrt{3} \\ &= \frac{8\sqrt{3}}{3}\end{aligned}$$

The exact area of triangle DEF is

$$\frac{8\sqrt{3}}{3}$$



7. **(a)** Using the identity $zz^* = |z|^2$, or otherwise, show that if w is any root of unity then

$$|w - (1+\mathrm{i})|^2 = 3 - (1-\mathrm{i})w - (1+\mathrm{i})w^* \qquad \textbf{[3]}$$

The diagram shows a regular decagon $A_1A_2\ldots A_{10}$ whose vertices all lie on the unit circle with centre at the origin O and A_1 at $(1,0)$. The point P lies in the same plane as the decagon and has coordinates $(1,1)$.

(b) Using the result from part (a), find

$$\sum_{r=1}^{5}(PA_{2r-1})^2 \qquad \textbf{[4]}$$

Solution

(a) Using

$$|z|^2 = zz^*$$

we have

$$\begin{aligned}|w-(1+\mathrm{i})|^2 &= \big(w-(1+\mathrm{i})\big)\big(w^*-(1-\mathrm{i})\big)\\ &= ww^* - (1-\mathrm{i})w - (1+\mathrm{i})w^* + (1+\mathrm{i})(1-\mathrm{i})\end{aligned}$$

Now w is a root of unity, so $|w| = 1$. Hence

$$ww^* = |w|^2 = 1$$

Also,

$$(1+\mathrm{i})(1-\mathrm{i}) = 1 - \mathrm{i}^2 = 2$$

So

$$\begin{aligned}|w-(1+\mathrm{i})|^2 &= 1 - (1-\mathrm{i})w - (1+\mathrm{i})w^* + 2\\ &= 3 - (1-\mathrm{i})w - (1+\mathrm{i})w^*\end{aligned}$$

Therefore

$$|w-(1+\mathrm{i})|^2 = 3 - (1-\mathrm{i})w - (1+\mathrm{i})w^*$$

(b) Let the complex number representing A_{2r-1} be w_r.
Since the decagon is regular and $A_1 = (1,0)$, the odd-numbered vertices are at arguments

$$0,\ \frac{2\pi}{5},\ \frac{4\pi}{5},\ \frac{6\pi}{5},\ \frac{8\pi}{5}$$

so

$$w_r = e^{2\pi\mathrm{i}(r-1)/5} \qquad (r = 1,2,3,4,5)$$

These are exactly the 5th roots of unity.

Also, the point $P = (1, 1)$ corresponds to the complex number $1 + \mathrm{i}$. Therefore

$$PA_{2r-1} = |w_r - (1 + \mathrm{i})|$$

and so

$$(PA_{2r-1})^2 = |w_r - (1 + \mathrm{i})|^2$$

Using the result from part (a),

$$(PA_{2r-1})^2 = 3 - (1 - \mathrm{i})w_r - (1 + \mathrm{i})w_r^*$$

Now sum for $r = 1$ to 5:

$$\begin{aligned}\sum_{r=1}^{5}(PA_{2r-1})^2 &= \sum_{r=1}^{5}(3 - (1 - \mathrm{i})w_r - (1 + \mathrm{i})w_r^*) \\ &= 15 - (1 - \mathrm{i})\sum_{r=1}^{5} w_r - (1 + \mathrm{i})\sum_{r=1}^{5} w_r^*\end{aligned}$$

Since w_1, w_2, w_3, w_4, w_5 are the 5th roots of unity,

$$\sum_{r=1}^{5} w_r = 0$$

Taking complex conjugates gives

$$\sum_{r=1}^{5} w_r^* = 0$$

Hence

$$\sum_{r=1}^{5}(PA_{2r-1})^2 = 15$$

8. You are given the complex number $\alpha = \cos\frac{3\pi}{5} + \mathrm{i}\sin\frac{3\pi}{5}$ and the equation $z^5 = -1$.

(a) Show that α is a root of the equation. **[2]**

(b) Write down the other four roots of the equation. **[1]**

(c) Show that

$$\alpha^4 - \alpha^3 + \alpha^2 - \alpha + 1 = 0$$ **[2]**

(d) Hence show that

$$\left(\alpha + \frac{1}{\alpha}\right)^2 - \left(\alpha + \frac{1}{\alpha}\right) - 1 = 0$$ **[3]**

(e) Hence determine the value of $\cos\frac{3\pi}{5}$ in the form $a + b\sqrt{c}$, where a, b and c are rational numbers to be found. **[4]**

Solution

(a) We have

$$\alpha = \cos\frac{3\pi}{5} + \mathrm{i}\sin\frac{3\pi}{5}$$

Using De Moivre's theorem,

$$\begin{aligned}\alpha^5 &= \cos\left(5\cdot\frac{3\pi}{5}\right) + \mathrm{i}\sin\left(5\cdot\frac{3\pi}{5}\right)\\ &= \cos 3\pi + \mathrm{i}\sin 3\pi\\ &= -1 + 0\mathrm{i}\\ &= -1\end{aligned}$$

So α satisfies $z^5 = -1$, and hence α is a root of the equation.

(b) Since

$$-1 = \cos\big((2k+1)\pi\big) + \mathrm{i}\sin\big((2k+1)\pi\big)$$

the five roots are

$$z = \cos\frac{(2k+1)\pi}{5} + \mathrm{i}\sin\frac{(2k+1)\pi}{5} \qquad (k = 0, 1, 2, 3, 4)$$

Given that $\alpha = \cos\frac{3\pi}{5} + \mathrm{i}\sin\frac{3\pi}{5}$, the other four roots are

$$\cos\frac{\pi}{5} + \mathrm{i}\sin\frac{\pi}{5}, \quad \cos\pi + \mathrm{i}\sin\pi, \quad \cos\frac{7\pi}{5} + \mathrm{i}\sin\frac{7\pi}{5}, \quad \cos\frac{9\pi}{5} + \mathrm{i}\sin\frac{9\pi}{5}$$

(c) From part (a), $\alpha^5 = -1$, so

$$\alpha^5 + 1 = 0$$

Divide $x^5 + 1$ by $x + 1$ to obtain the factorisation

$$x^5 + 1 = (x+1)(x^4 - x^3 + x^2 - x + 1)$$

Hence

$$(\alpha + 1)(\alpha^4 - \alpha^3 + \alpha^2 - \alpha + 1) = 0$$

But $\alpha \neq -1$, since $\alpha = \cos\frac{3\pi}{5} + \mathrm{i}\sin\frac{3\pi}{5}$ and $\sin\frac{3\pi}{5} \neq 0$.
Therefore

$$\alpha^4 - \alpha^3 + \alpha^2 - \alpha + 1 = 0$$

(d) Since $\alpha \neq 0$, divide the result from part (c) by α^2:

$$\begin{aligned}\frac{\alpha^4}{\alpha^2} - \frac{\alpha^3}{\alpha^2} + \frac{\alpha^2}{\alpha^2} - \frac{\alpha}{\alpha^2} + \frac{1}{\alpha^2} &= 0\\ \alpha^2 - \alpha + 1 - \frac{1}{\alpha} + \frac{1}{\alpha^2} &= 0\end{aligned}$$

Let

$$t = \alpha + \frac{1}{\alpha}$$

Then

$$\begin{aligned}t^2 &= \left(\alpha + \frac{1}{\alpha}\right)^2\\ &= \alpha^2 + 2 + \frac{1}{\alpha^2}\end{aligned}$$

so

$$\alpha^2 + \frac{1}{\alpha^2} = t^2 - 2$$

Also,

$$\alpha + \frac{1}{\alpha} = t$$

Substituting into

$$\alpha^2 - \alpha + 1 - \frac{1}{\alpha} + \frac{1}{\alpha^2} = 0$$

gives

$$\begin{aligned}\left(\alpha^2 + \frac{1}{\alpha^2}\right) - \left(\alpha + \frac{1}{\alpha}\right) + 1 &= 0\\ (t^2 - 2) - t + 1 &= 0\\ t^2 - t - 1 &= 0\end{aligned}$$

Therefore,

$$\left(\alpha + \frac{1}{\alpha}\right)^2 - \left(\alpha + \frac{1}{\alpha}\right) - 1 = 0$$

(e) From part (d), let

$$t = \alpha + \frac{1}{\alpha}$$

so that

$$t^2 - t - 1 = 0$$

Solving this quadratic,

$$t = \frac{1 \pm \sqrt{5}}{2}$$

Now

$$\alpha = \cos\frac{3\pi}{5} + \mathrm{i}\sin\frac{3\pi}{5}$$

and since $|\alpha| = 1$,

$$\frac{1}{\alpha} = \cos\frac{3\pi}{5} - \mathrm{i}\sin\frac{3\pi}{5}$$

Hence

$$\begin{aligned}\alpha + \frac{1}{\alpha} &= \left(\cos\frac{3\pi}{5} + \mathrm{i}\sin\frac{3\pi}{5}\right) + \left(\cos\frac{3\pi}{5} - \mathrm{i}\sin\frac{3\pi}{5}\right)\\ &= 2\cos\frac{3\pi}{5}\end{aligned}$$

So

$$2\cos\frac{3\pi}{5} = \frac{1 \pm \sqrt{5}}{2}$$

Therefore

$$\cos\frac{3\pi}{5} = \frac{1 \pm \sqrt{5}}{4}$$

Since $\frac{3\pi}{5} = 108^\circ$ is in quadrant II, $\cos\frac{3\pi}{5} < 0$, so we take the negative value:

$$\cos\frac{3\pi}{5} = \frac{1 - \sqrt{5}}{4}$$

So the required value is

$$\cos\frac{3\pi}{5} = \frac{1 - \sqrt{5}}{4}$$

9. The points representing the complex numbers $z_1 = 6 + 11\mathrm{i}$ and $z_2 = -8 - 3\mathrm{i}$ are opposite vertices of a regular octagon, K, in the complex plane.
The centre of K represents the complex number α.

(a) Show that $\alpha = -1 + 4\mathrm{i}$. **[2]**

Given that

$$\beta = \frac{1 - \mathrm{i}}{14}$$

(b) Show that

$$\beta(z_1 - \alpha) = 1$$ **[2]**

The vertices of K are given by the roots of the equation

$$(\beta(z - \alpha))^8 = 1$$

(c) (i) Write down the roots of the equation $w^8 = 1$ in the form $re^{\mathrm{i}\theta}$. **[1]**

(ii) Hence, or otherwise, determine the positions of the other six vertices of K, giving your answers as complex numbers in Cartesian form. **[4]**

Solution

(a) Since z_1 and z_2 are opposite vertices of the regular octagon, the centre is the midpoint of the line joining them.

$$\alpha = \frac{z_1 + z_2}{2} = \frac{(6 + 11\mathrm{i}) + (-8 - 3\mathrm{i})}{2}$$

$$\alpha = \frac{-2 + 8\mathrm{i}}{2} = -1 + 4\mathrm{i}$$

So $\alpha = -1 + 4\mathrm{i}$.

(b) First find $z_1 - \alpha$:

$$z_1 - \alpha = (6 + 11\mathrm{i}) - (-1 + 4\mathrm{i}) = 7 + 7\mathrm{i} = 7(1 + \mathrm{i})$$

Now multiply by β:

$$\beta(z_1 - \alpha) = \frac{1 - \mathrm{i}}{14} \cdot 7(1 + \mathrm{i}) = \frac{(1 - \mathrm{i})(1 + \mathrm{i})}{2}$$

Using $(1 - \mathrm{i})(1 + \mathrm{i}) = 1 - \mathrm{i}^2 = 2$,

$$\beta(z_1 - \alpha) = \frac{2}{2} = 1$$

Hence $\beta(z_1 - \alpha) = 1$.

(c) (i) The roots of $w^8 = 1$ are the eighth roots of unity:

$$w = e^{\mathrm{i}k\pi/4}, \qquad k = 0, 1, 2, 3, 4, 5, 6, 7$$

So in the form $re^{\mathrm{i}\theta}$, we have $r = 1$ and $\theta = \dfrac{k\pi}{4}$.

(ii) From

$$(\beta(z - \alpha))^8 = 1$$

let

$$w = \beta(z - \alpha)$$

Then $w^8 = 1$, so

$$z - \alpha = \frac{w}{\beta} \qquad \Rightarrow \qquad z = \alpha + \frac{w}{\beta}$$

Now

$$\frac{1}{\beta} = \frac{14}{1-\mathrm{i}} = \frac{14(1+\mathrm{i})}{(1-\mathrm{i})(1+\mathrm{i})} = \frac{14(1+\mathrm{i})}{2} = 7(1+\mathrm{i})$$

Hence

$$z = \alpha + 7(1+\mathrm{i})w$$

Using $\alpha = -1 + 4\mathrm{i}$,

$$z = -1 + 4\mathrm{i} + 7(1+\mathrm{i})w$$

Now substitute the eight roots of $w^8 = 1$.
For $w = 1$,

$$z = -1 + 4\mathrm{i} + 7(1+\mathrm{i}) = 6 + 11\mathrm{i}$$

which is z_1.
For $w = -1$,

$$z = -1 + 4\mathrm{i} - 7(1+\mathrm{i}) = -8 - 3\mathrm{i}$$

which is z_2.
So the other six vertices come from the remaining six values of w.
For $w = e^{\mathrm{i}\pi/4} = \dfrac{1+\mathrm{i}}{\sqrt{2}}$,

$$z = -1 + 4\mathrm{i} + 7(1+\mathrm{i})\frac{1+\mathrm{i}}{\sqrt{2}} = -1 + 4\mathrm{i} + 7\sqrt{2}\,\mathrm{i}$$

so

$$z = -1 + (4 + 7\sqrt{2})\mathrm{i}$$

For $w = e^{\mathrm{i}\pi/2} = \mathrm{i}$,

$$z = -1 + 4\mathrm{i} + 7(1+\mathrm{i})\mathrm{i} = -1 + 4\mathrm{i} + 7(-1+\mathrm{i})$$

so

$$z = -8 + 11\mathrm{i}$$

For $w = e^{3\mathrm{i}\pi/4} = \dfrac{-1+\mathrm{i}}{\sqrt{2}}$,

$$z = -1 + 4\mathrm{i} + 7(1+\mathrm{i})\frac{-1+\mathrm{i}}{\sqrt{2}} = -1 + 4\mathrm{i} - 7\sqrt{2}$$

so

$$z = (-1 - 7\sqrt{2}) + 4\mathrm{i}$$

For $w = e^{5\mathrm{i}\pi/4} = \dfrac{-1-\mathrm{i}}{\sqrt{2}}$,

$$z = -1 + 4\mathrm{i} + 7(1+\mathrm{i})\frac{-1-\mathrm{i}}{\sqrt{2}} = -1 + 4\mathrm{i} - 7\sqrt{2}\,\mathrm{i}$$

so

$$z = -1 + (4 - 7\sqrt{2})\mathrm{i}$$

For $w = e^{3\mathrm{i}\pi/2} = -\mathrm{i}$,

$$z = -1 + 4\mathrm{i} + 7(1+\mathrm{i})(-\mathrm{i}) = -1 + 4\mathrm{i} + 7(1-\mathrm{i})$$

so

$$z = 6 - 3\mathrm{i}$$

For $w = e^{7\mathrm{i}\pi/4} = \dfrac{1-\mathrm{i}}{\sqrt{2}}$,

$$z = -1 + 4\mathrm{i} + 7(1+\mathrm{i})\frac{1-\mathrm{i}}{\sqrt{2}} = -1 + 4\mathrm{i} + 7\sqrt{2}$$

so

$$z = (-1 + 7\sqrt{2}) + 4\mathrm{i}$$

Therefore the other six vertices are

$$\begin{gathered} -1 + (4 + 7\sqrt{2})\mathrm{i}, \quad -8 + 11\mathrm{i}, \quad (-1 - 7\sqrt{2}) + 4\mathrm{i}, \\ -1 + (4 - 7\sqrt{2})\mathrm{i}, \quad 6 - 3\mathrm{i}, \quad (-1 + 7\sqrt{2}) + 4\mathrm{i} \end{gathered}$$

10. In an Argand diagram, the points A and B are represented by the complex numbers $1+4\mathrm{i}$ and $7+2\mathrm{i}$ respectively. The circle C passes through A and B, and its centre lies on the real axis.

(a) Find the equation of C, giving your answer in the form

$$|z-a| = b \qquad a \in \mathbb{C},\ b \in \mathbb{R}$$ **[4]**

The circle D, with equation $|z-(1+\mathrm{i})| = 5$, intersects C at the points representing the complex numbers z_1 and z_2.

(b) Find the complex numbers z_1 and z_2. **[5]**

Solution

(a) Let the centre of circle C be the point represented by the real number h, since the centre lies on the real axis.

So the centre is $(h, 0)$ on the Argand diagram.

Since $A(1,4)$ and $B(7,2)$ both lie on C, their distances from the centre are equal:

$$\begin{aligned}(h-1)^2 + 4^2 &= (h-7)^2 + 2^2\\ (h-1)^2 + 16 &= (h-7)^2 + 4\end{aligned}$$

Expanding:

$$\begin{aligned}h^2 - 2h + 1 + 16 &= h^2 - 14h + 49 + 4\\ -2h + 17 &= -14h + 53\\ 12h &= 36\\ h &= 3\end{aligned}$$

So the centre is the complex number 3.

Now find the radius using point A:

$$\begin{aligned}r^2 &= (3-1)^2 + 4^2\\ &= 2^2 + 4^2\\ &= 4 + 16\\ &= 20\end{aligned}$$

Hence

$$r = \sqrt{20} = 2\sqrt{5}$$

Therefore the equation of C is

$$|z-3| = 2\sqrt{5}$$

(b) Let

$$z = x + \mathrm{i}y$$

Since z lies on C, we have

$$|z-3| = 2\sqrt{5}$$

so

$$(x-3)^2 + y^2 = 20$$

Since z also lies on D, whose equation is $|z-(1+\mathrm{i})| = 5$, we have

$$(x-1)^2 + (y-1)^2 = 25$$

So the points of intersection satisfy the simultaneous equations

$$\begin{aligned}(x-3)^2+y^2&=20\\(x-1)^2+(y-1)^2&=25\end{aligned}$$

Expand the second equation:

$$\begin{aligned}x^2-2x+1+y^2-2y+1&=25\\x^2+y^2-2x-2y-23&=0\end{aligned}$$

Expand the first equation:

$$\begin{aligned}x^2-6x+9+y^2&=20\\x^2+y^2-6x-11&=0\end{aligned}$$

Subtract the first from the second:

$$\begin{aligned}(x^2+y^2-2x-2y-23)-(x^2+y^2-6x-11)&=0\\-2x-2y-23+6x+11&=0\\4x-2y-12&=0\\2x-y-6&=0\end{aligned}$$

Hence

$$y=2x-6$$

Substitute into $(x-3)^2+y^2=20$:

$$\begin{aligned}(x-3)^2+(2x-6)^2&=20\\(x-3)^2+4(x-3)^2&=20\\5(x-3)^2&=20\\(x-3)^2&=4\end{aligned}$$

So

$$x-3=\pm 2$$

giving

$$x=1 \quad \text{or} \quad x=5$$

Now use $y=2x-6$.
If $x=1$,

$$y=2(1)-6=-4$$

If $x=5$,

$$y=2(5)-6=4$$

Therefore the two intersection points are represented by

$$z_1=1-4\mathrm{i}, \qquad z_2=5+4\mathrm{i}$$

11. The cubic equation

$$z^3 + pz^2 + qz + 24 = 0$$

where p and q are real, has roots α, β and γ.
It is given that $\alpha = -2 + 2\mathrm{i}$.

(a) (i) Write down another root, β, of the equation. **[1]**
(ii) Find the third root, γ. **[3]**
(iii) Find the values of p and q. **[3]**

(b) (i) Express α in the form $re^{\mathrm{i}\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. **[2]**
(ii) Hence show that **[2]**

$$(-2 + 2\mathrm{i})^n = (2\sqrt{2})^n \left(\cos\frac{3n\pi}{4} + \mathrm{i}\sin\frac{3n\pi}{4}\right)$$

(iii) Hence show that

$$\alpha^n + \beta^n + \gamma^n = 2(2\sqrt{2})^n \cos\frac{3n\pi}{4} + (-3)^n$$

where n is an integer. **[3]**

Solution

(a) (i) Since the coefficients of the cubic are real, complex roots occur in conjugate pairs.
So the other complex root is

$$\beta = -2 - 2\mathrm{i}$$

(ii) For the monic cubic

$$z^3 + pz^2 + qz + 24 = 0$$

the product of the roots is

$$\alpha\beta\gamma = -24$$

First find $\alpha\beta$:

$$\begin{aligned}\alpha\beta &= (-2 + 2\mathrm{i})(-2 - 2\mathrm{i}) \\ &= (-2)^2 - (2\mathrm{i})^2 \\ &= 4 - (-4) \\ &= 8\end{aligned}$$

So

$$8\gamma = -24$$

and hence

$$\gamma = -3$$

(iii) Using the relationships between roots and coefficients:

$$\alpha + \beta + \gamma = -p$$

Now

$$\begin{aligned}\alpha + \beta + \gamma &= (-2 + 2\mathrm{i}) + (-2 - 2\mathrm{i}) + (-3) \\ &= -7\end{aligned}$$

so

$$-p = -7$$

giving

$$p = 7$$

Also

$$\alpha\beta + \beta\gamma + \gamma\alpha = q$$

We already know that $\alpha\beta = 8$, and

$$\begin{aligned}\beta\gamma + \gamma\alpha &= \gamma(\alpha+\beta)\\ &= (-3)\big((-2+2\mathrm{i}) + (-2-2\mathrm{i})\big)\\ &= (-3)(-4)\\ &= 12\end{aligned}$$

Therefore

$$q = 8 + 12 = 20$$

So the values are

$$p = 7, \qquad q = 20$$

(b) (i) For

$$\alpha = -2 + 2\mathrm{i}$$

its modulus is

$$\begin{aligned}r &= |\alpha|\\ &= \sqrt{(-2)^2 + 2^2}\\ &= \sqrt{8}\\ &= 2\sqrt{2}\end{aligned}$$

The point $(-2, 2)$ lies in the second quadrant, so the principal argument is

$$\theta = \frac{3\pi}{4}$$

Hence

$$\alpha = 2\sqrt{2}\, e^{\mathrm{i}3\pi/4}$$

(ii) From part (i),

$$\alpha = 2\sqrt{2}\, e^{\mathrm{i}3\pi/4}$$

so

$$\begin{aligned}\alpha^n &= \left(2\sqrt{2}\, e^{\mathrm{i}3\pi/4}\right)^n\\ &= (2\sqrt{2})^n e^{\mathrm{i}3n\pi/4}\end{aligned}$$

Using $e^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$,

$$(-2+2\mathrm{i})^n = (2\sqrt{2})^n \left(\cos\frac{3n\pi}{4} + \mathrm{i}\sin\frac{3n\pi}{4}\right)$$

(iii) Since

$$\beta = -2 - 2\mathrm{i} = 2\sqrt{2}\, e^{-\mathrm{i}3\pi/4}$$

we have

$$\begin{aligned}\beta^n &= \left(2\sqrt{2}\, e^{-\mathrm{i}3\pi/4}\right)^n\\ &= (2\sqrt{2})^n \left(\cos\left(-\frac{3n\pi}{4}\right) + \mathrm{i}\sin\left(-\frac{3n\pi}{4}\right)\right)\\ &= (2\sqrt{2})^n \left(\cos\frac{3n\pi}{4} - \mathrm{i}\sin\frac{3n\pi}{4}\right)\end{aligned}$$

Therefore

$$\begin{aligned}\alpha^n + \beta^n &= (2\sqrt{2})^n \left(\cos\frac{3n\pi}{4} + \mathrm{i}\sin\frac{3n\pi}{4}\right) + (2\sqrt{2})^n \left(\cos\frac{3n\pi}{4} - \mathrm{i}\sin\frac{3n\pi}{4}\right)\\ &= 2(2\sqrt{2})^n \cos\frac{3n\pi}{4}\end{aligned}$$

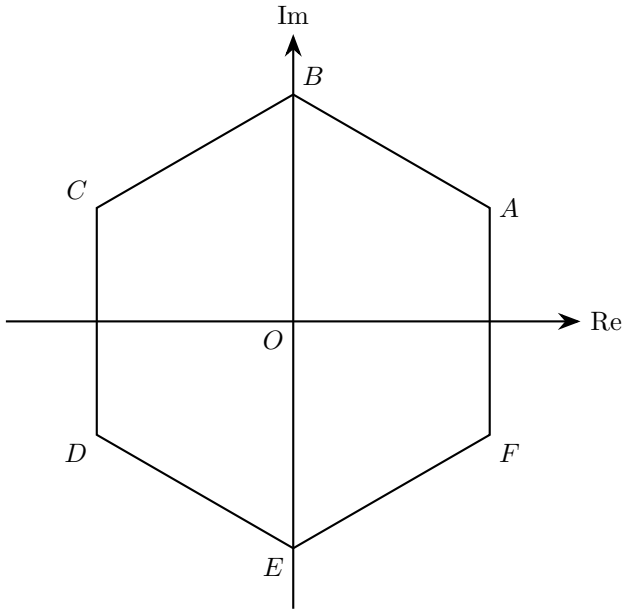
Also, from part (a), $\gamma = -3$, so

$$\gamma^n = (-3)^n$$

Hence

$$\alpha^n + \beta^n + \gamma^n = 2(2\sqrt{2})^n \cos\frac{3n\pi}{4} + (-3)^n$$

as required.



12. In the diagram, the points A, B, C, D, E and F represent the complex sixth roots of -729 on an Argand diagram. The centroids of triangles OAB, OBC, OCD, ODE, OEF and OFA are G, H, I, J, K and L respectively.

(a) Write down, in exponential form, the complex numbers represented by the points A, B, C, D, E and F. **[2]**

(b) When these complex numbers are multiplied by the complex number w, the resulting complex numbers are represented by the points G, H, I, J, K and L.
Find w in exponential form. **[4]**

(c) You are given that G, H, I, J, K and L represent roots of the equation $z^6 = p$.
Find p. **[2]**

Solution

(a) Since

$$-729 = 729e^{\mathrm{i}\pi} = 3^6 e^{\mathrm{i}(\pi+2k\pi)}$$

the sixth roots are

$$z = 3e^{\mathrm{i}\left(\frac{\pi+2k\pi}{6}\right)} = 3e^{\mathrm{i}\left(\frac{\pi}{6}+\frac{k\pi}{3}\right)}, \qquad k = 0, 1, 2, 3, 4, 5$$

Taking these in order round the Argand diagram gives

$$A = 3e^{\mathrm{i}\pi/6}, \qquad B = 3e^{\mathrm{i}\pi/2}, \qquad C = 3e^{\mathrm{i}5\pi/6},$$
$$D = 3e^{\mathrm{i}7\pi/6}, \qquad E = 3e^{\mathrm{i}3\pi/2}, \qquad F = 3e^{\mathrm{i}11\pi/6}$$

(b) The centroid of triangle OAB is

$$G = \frac{O+A+B}{3} = \frac{A+B}{3}$$

If multiplication by w maps A to G, then

$$Aw = G = \frac{A+B}{3}$$

so

$$w = \frac{A+B}{3A}$$

Now B is the next sixth root after A, so

$$B = Ae^{\mathrm{i}\pi/3}$$

Hence

$$\begin{aligned} w &= \frac{A + Ae^{\mathrm{i}\pi/3}}{3A} \\ &= \frac{1 + e^{\mathrm{i}\pi/3}}{3} \end{aligned}$$

Now

$$e^{\mathrm{i}\pi/3} = \frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$$

so

$$1 + e^{\mathrm{i}\pi/3} = \frac{3}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$$

Its modulus is

$$\sqrt{\left(\frac{3}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = \sqrt{\frac{9}{4} + \frac{3}{4}} = \sqrt{3}$$

and its argument is

$$\tan^{-1}\left(\frac{\sqrt{3}/2}{3/2}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$$

So

$$1 + e^{\mathrm{i}\pi/3} = \sqrt{3}\, e^{\mathrm{i}\pi/6}$$

Therefore

$$w = \frac{\sqrt{3}}{3}e^{\mathrm{i}\pi/6} = \frac{1}{\sqrt{3}}e^{\mathrm{i}\pi/6}$$

Also, for example,

$$H = \frac{B + C}{3} = \frac{B + Be^{\mathrm{i}\pi/3}}{3} = Bw$$

and similarly for the other consecutive pairs, so this same w maps A, B, C, D, E, F to G, H, I, J, K, L. Hence

$$w = \frac{1}{\sqrt{3}}e^{\mathrm{i}\pi/6}$$

(c) Let r be any sixth root of -729. Then the corresponding point among G, H, I, J, K, L is

$$z = wr$$

So

$$z^6 = w^6 r^6 = w^6(-729)$$

Now

$$\begin{aligned} w^6 &= \left(\frac{1}{\sqrt{3}}e^{\mathrm{i}\pi/6}\right)^6 \\ &= \left(\frac{1}{\sqrt{3}}\right)^6 e^{\mathrm{i}\pi} \\ &= \frac{1}{27}(-1) \\ &= -\frac{1}{27} \end{aligned}$$

Hence

$$p = w^6(-729) = \left(-\frac{1}{27}\right)(-729) = 27$$

Therefore

$$p = 27$$

13. (i) Write down, in any form, all the complex roots of the equation

$$w^{11} = -1$$ **[2]**

(ii) Explain why the equation

$$(z+1)^{11} = -(z-1)^{11}$$

has exactly 10 non-real roots and show that they may be expressed in the form

$$-\mathrm{i}\cot\left(\frac{m\pi}{22}\right)$$ **[6]**

where $m = \pm 1, \pm 3, \pm 5, \pm 7, \pm 9$.

(iii) Show that

$$\left\{-\mathrm{i}\cot\left(\frac{m\pi}{22}\right)\right\}\left\{\mathrm{i}\cot\left(\frac{m\pi}{22}\right)\right\} = \cot^2\left(\frac{m\pi}{22}\right)$$ **[1]**

(iv) Given that the product of the non-zero roots of the equation in part (ii) is 11, find the value of

$$\tan^2\left(\frac{\pi}{22}\right)\tan^2\left(\frac{3\pi}{22}\right)\tan^2\left(\frac{5\pi}{22}\right)\tan^2\left(\frac{7\pi}{22}\right)\tan^2\left(\frac{9\pi}{22}\right)$$ **[2]**

Solution

(i) Write -1 in exponential form as

$$-1 = e^{\mathrm{i}(2k+1)\pi} \qquad (k \in \mathbb{Z})$$

So if $w^{11} = -1$, then

$$w = e^{\mathrm{i}(2k+1)\pi/11}$$

Taking the 11 distinct values gives

$$w = e^{\mathrm{i}(2k+1)\pi/11}, \qquad k = 0, 1, \ldots, 10$$

So all the complex roots are

$$e^{\mathrm{i}\pi/11},\ e^{3\mathrm{i}\pi/11},\ e^{5\mathrm{i}\pi/11},\ \ldots,\ e^{21\mathrm{i}\pi/11}$$

(ii) First note that $z = 1$ is not a root of

$$(z+1)^{11} = -(z-1)^{11}$$

because this would give $2^{11} = 0$, which is false.
So we may divide by $(z-1)^{11}$:

$$\left(\frac{z+1}{z-1}\right)^{11} = -1$$

Let

$$w = \frac{z+1}{z-1}$$

Then

$$w^{11} = -1$$

From part (i), the 11 possible values of w are

$$w = e^{\mathrm{i}\theta}, \qquad \theta = \frac{(2k+1)\pi}{11}, \qquad k = 0, 1, \ldots, 10$$

Now solve for z in terms of w:

$$\begin{aligned} w &= \frac{z+1}{z-1} \\ w(z-1) &= z+1 \\ wz - w &= z+1 \\ z(w-1) &= w+1 \\ z &= \frac{w+1}{w-1} \end{aligned}$$

So

$$z = \frac{e^{\mathrm{i}\theta}+1}{e^{\mathrm{i}\theta}-1}$$

Now multiply top and bottom by $e^{-\mathrm{i}\theta/2}$:

$$\begin{aligned} z &= \frac{e^{\mathrm{i}\theta/2}+e^{-\mathrm{i}\theta/2}}{e^{\mathrm{i}\theta/2}-e^{-\mathrm{i}\theta/2}} \\ &= \frac{2\cos(\theta/2)}{2\mathrm{i}\sin(\theta/2)} \\ &= -\mathrm{i}\cot\left(\frac{\theta}{2}\right) \end{aligned}$$

Since $\theta = \dfrac{(2k+1)\pi}{11}$, this gives

$$z = -\mathrm{i}\cot\left(\frac{(2k+1)\pi}{22}\right) \qquad k = 0, 1, \ldots, 10$$

So the 11 roots are obtained from

$$m = 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21$$

in

$$z = -\mathrm{i}\cot\left(\frac{m\pi}{22}\right)$$

When $m = 11$,

$$z = -\mathrm{i}\cot\left(\frac{11\pi}{22}\right) = -\mathrm{i}\cot\left(\frac{\pi}{2}\right) = 0$$

so this is the one real root.

To see there are no other real roots, suppose z is real. Then

$$w = \frac{z+1}{z-1}$$

is also real. But the only real solution of $w^{11} = -1$ is $w = -1$, which gives

$$z = \frac{-1+1}{-1-1} = 0$$

Hence there is exactly one real root, namely 0, so the remaining $11 - 1 = 10$ roots are non-real.

Now rewrite the values with $m = 13, 15, 17, 19, 21$. If $r = 1, 3, 5, 7, 9$, then

$$\begin{aligned} -\mathrm{i}\cot\left(\frac{(22-r)\pi}{22}\right) &= -\mathrm{i}\cot\left(\pi - \frac{r\pi}{22}\right) \\ &= -\mathrm{i}\left(-\cot\left(\frac{r\pi}{22}\right)\right) \\ &= \mathrm{i}\cot\left(\frac{r\pi}{22}\right) \\ &= -\mathrm{i}\cot\left(-\frac{r\pi}{22}\right) \end{aligned}$$

Therefore the 10 non-real roots may be written as

$$-\mathrm{i}\cot\left(\frac{m\pi}{22}\right), \qquad m = \pm 1, \pm 3, \pm 5, \pm 7, \pm 9$$

(iii) Let

$$\alpha = \frac{m\pi}{22}$$

Then

$$\begin{aligned}(-\mathrm{i}\cot\alpha)(\mathrm{i}\cot\alpha) &= (-\mathrm{i}\cdot\mathrm{i})\cot^2\alpha \\ &= -\mathrm{i}^2\cot^2\alpha \\ &= \cot^2\alpha\end{aligned}$$

So

$$\left\{-\mathrm{i}\cot\left(\frac{m\pi}{22}\right)\right\}\left\{\mathrm{i}\cot\left(\frac{m\pi}{22}\right)\right\} = \cot^2\left(\frac{m\pi}{22}\right)$$

(iv) The non-zero roots occur in pairs

$$-\mathrm{i}\cot\left(\frac{m\pi}{22}\right) \quad \text{and} \quad -\mathrm{i}\cot\left(-\frac{m\pi}{22}\right) = \mathrm{i}\cot\left(\frac{m\pi}{22}\right)$$

for

$$m = 1, 3, 5, 7, 9$$

Given that the product of the non-zero roots is 11,

$$\begin{aligned}11 &= \prod_{m=1,3,5,7,9}\left\{-\mathrm{i}\cot\left(\frac{m\pi}{22}\right)\right\}\left\{\mathrm{i}\cot\left(\frac{m\pi}{22}\right)\right\} \\ &= \prod_{m=1,3,5,7,9}\cot^2\left(\frac{m\pi}{22}\right)\end{aligned}$$

Therefore

$$\cot^2\left(\frac{\pi}{22}\right)\cot^2\left(\frac{3\pi}{22}\right)\cot^2\left(\frac{5\pi}{22}\right)\cot^2\left(\frac{7\pi}{22}\right)\cot^2\left(\frac{9\pi}{22}\right) = 11$$

Taking reciprocals,

$$\tan^2\left(\frac{\pi}{22}\right)\tan^2\left(\frac{3\pi}{22}\right)\tan^2\left(\frac{5\pi}{22}\right)\tan^2\left(\frac{7\pi}{22}\right)\tan^2\left(\frac{9\pi}{22}\right) = \frac{1}{11}$$

So the required value is

$$\frac{1}{11}$$